

INDU211 Introduction to Production and Manufacturing Systems Fall 2026	
Course Instructor: Masoumeh Kazemi Zanjani, Eng., Ph.D. Office: EV04-153 Email: masoumeh.kazemizanjani@concordia.ca	
Office Hours: <i>Wednesday, 1:00-3:00 pm, EV04-153</i>	
Tutorials: Please see your class schedule for details Tutor: Fatemeh Safaei, Email: fatemeh.safaei@mail.concordia.ca	
Labs: N/A	
Course Calendar Description: History of industrial engineering; Role of industrial engineers; Types of manufacturing and production systems; Job design and work measurement; Quality control and management; Introduction to solution methodologies for problems which relate to the design and operation of integrated production systems of humans, machines, information, and materials.	
Prerequisites: N/A Co-requisites: N/A	
Specific Knowledge and Skills Needed for this Course: Students taking this course are expected to have sufficient knowledge of the following topics. Should you have difficulties in any of these topics, you are strongly encouraged to review them before the DNE deadline. • <i>Principles of natural science and social science</i> • <i>Fundamentals of calculus, linear algebra and probability/statistics</i>	
Course materials • Introduction to Industrial and Systems Engineering, W.C. Turner, J.H. Mize, K.E. Case, J.W. Nazemetz, Third Edition, Prentice Hall, 1993, ISBN: 0-13-481789-3 (Off-print, required chapters available on course Moodle site) • Reference Book: Introduction to Industrial and Systems Engineering, by Robert Wayne Atkins, Grandpappy Inc, 2019, ISBN-13 : 978-1732788312 • Instructor's lecture notes to be posted on course Moodle site Other materials: • Engineers Canada Documents (https://engineerscanada.ca/regulatory-excellence/national-engineering-guidelines) • IISE website materials (https://www.iise.org/details.aspx?id=282)	

Grading Scheme

- Assignment: 0%
- Midterm Exam: 35%
- Final Exam: 50%
- Term Paper 15%

Passing Criteria: 50% or higher on the final exam mark AND the total mark.

“R” Grade: R grade will be assigned if the total mark is less than 25%.

Conversion from Numerical Marks to Alphabetical Grades (Subject to Revisions):

0-49	50-54	55-58	59-62	63-66	67-70	71-74	75-78	79-82	83-86	87-90	91-100
F	D	D+	C-	C	C+	B-	B	B+	A-	A	A+

Tentative Course Schedule

- Introduction to Engineering and Industrial Engineering (Weeks 1)
- Facilities Location and Layout, Material Handling, Distribution, and Routing (Weeks 2-3)
- Operations Planning and Control (Weeks 4-5)
- Midterm (Week 6)
- Deterministic Operations Research Models (Week 7)
- Probabilistic Operations Research Models (Week 8)
- Quality Control and Improvement (Weeks 9-10)
- Human Factors and Related Issues (Week 11)
- Project Management (Week 12)

Lab Details:

N/A

Engineering Tools:

N/A

Details on assessment tools:

N/A

Other information

- Term Paper Topic: You must submit a term paper on the topic suggested by the instructor.
- Term Paper Team: You are expected to work on the problem in a group with up to 6 members. Details are provide on the Term Paper Description Document.

Graduate Attributes:

The following is the list of graduate attributes (skills) that students use, learn and/or apply throughout the term.

- Knowledge base for engineering, Indicator:
 - Knowledge base in specific domain
- Individual and team work, Indicator:
 - Cooperation and work ethics;
 - Contribution: practical/conceptual;

○ Initiative and leadership • Life-long learning, Indicator: ○ Identifying missing knowledge and learning opportunities
Course Learning Outcomes (CLOs): By the end of this semester, students are expected to master the following Industrial Engineering concepts. • Recognize various decision-making problems arising in manufacturing and service systems • Define several concepts, tools, and techniques related to Industrial Engineering • Apply simple heuristics to solve small decision models arising in facility design, material handling production planning, among others • Demonstrate skills in individual and team work by working on a group term paper/project
Use of Concordia email Your Concordia email is the official channel for all university communications, as required by the university's Policy on Concordia email accounts for students and employees. Please use your Concordia email for all communication related to this course and check it regularly so you don't miss important information. Learn more about how to access your Concordia email.
Health and Safety Guidelines All health and safety rules specific to this course can be found in the lab manual. General health and safety instructions and available health and safety trainings are discussed at: Safety Programs - Concordia University
ON CAMPUS RESOURCES Student may find the full list of available resources at: • Important Services & Resources - Concordia University • A-Z Student Services - Concordia University While they are included in above mentioned two links, following list includes frequently searched services and resources available for students.
1. Academic Resources - Concordia University
2. Health & Wellness - Concordia University
3. Safety & security - Concordia University
4. International students - Concordia University